

Course 07: App Building

Comprehensive Learning Guide & Practical Prompting Playbook

MODULE OVERVIEW

Leverage AI as a full-stack pair programmer. Learn how to architect application schemas, design clean REST/GraphQL APIs, build modular UI components, and write maintainable code following software engineering best practices.

KEY LEARNING TOPICS

- System Architecture & Database Schema Design
- Full-Stack API Design & Route Handling (Node/Express, Python)
- Modular Frontend Component Architecture (React/Next.js)
- Debugging, Refactoring, & Writing Test Suites

PROMPTING STRATEGY & ARCHITECTURE

Specify stack, directory structures, exact component interfaces, state management patterns, and coding conventions (e.g., TypeScript, clean code principles).

ROLE

You are a Principal Full-Stack Engineer specializing in React, Node.js, and MongoDB.

TASK

Build a full modular REST API controller and React frontend component for a venue booking system ('HallHub').

TECH STACK

MongoDB (Mongoose), Node.js (Express), React (Hooks, Tailwind CSS).

INSTRUCTIONS

1. Define the Mongoose schema for `Venue` (name, capacity, price_per_hour, location, features).
2. Write an Express controller endpoint `POST /api/venues` with input validation.
3. Write a clean React component `VenueForm.jsx` that submits data to this endpoint with loading and error handling states.

CONSTRAINTS

- Follow clean architecture principles.
- Include proper try/catch error handling and HTTP status codes.

BEST PRACTICES & PRACTICAL TIPS

- Break app building into stages: Schema Design -> Backend API -> Frontend Components -> Integration.
- Ask AI to provide TypeScript interfaces or JSDoc comments to enforce structural clarity.
- Provide existing helper functions or utility setups so AI generates code that seamlessly fits your codebase.